

EmissionGate

Carbon at merge time. An agent that puts the kgCO2e of cloud and AI infrastructure into the pull request — and leaves the decision to a human.

EmissionGate — carbon check

 Acknowledgement required

+683.5

kgCO2e per year · range 263–1,104 (assumed load)

gpu_fleet: +2 × g5.2xlarge

Suggestion: weekday schedule (plan ✓)

Carbon is committed at merge and seen months later



Too late

Cloud carbon reports arrive monthly, by service and region — never per resource, never per change.



Wrong ranking

Cost tools optimise a different list: archiving 40 TB of logs saves money and ≈ 0 kgCO₂e; Graviton is the reverse.



Waste runs for months

An idle staging GPU fleet emits 401 kgCO₂e a year before anyone links it to the PR that created it.

Where the decision is made vs where carbon appears



PR merged

the decision



Deployed, runs 24 × 7

emissions start



Month-end bill

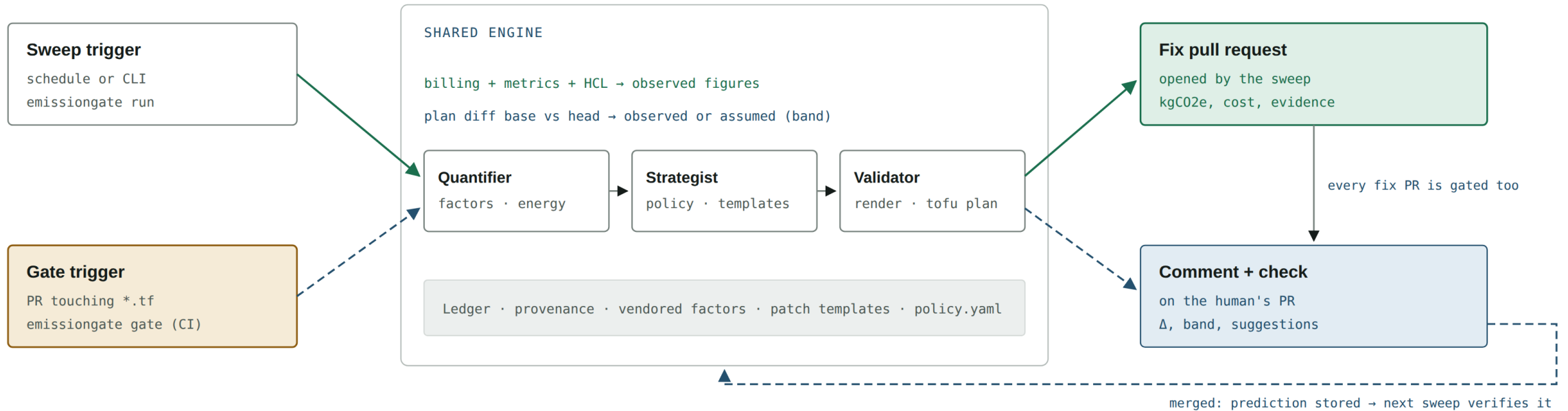
cost visible



Carbon report

carbon visible, by region

Two modes, one engine



Sweep

Finds waste in running infrastructure and drafts the fix as a PR, with kgCO2e, cost and evidence.

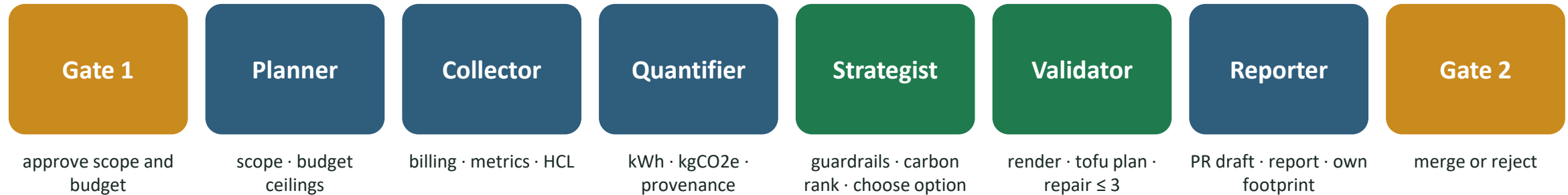
Gate

Checks every infrastructure PR before merge; posts the delta and validated lower-carbon suggestions. A repo opts in by adding one workflow file.

One set of numbers

Both modes share every component that produces a figure, so a comment and a report always agree.

How it works: deterministic numbers, bounded judgement



Deterministic core — every published number

Energy model, vendored factors, policy, ranking, patch templates, ledger, SCI.

It cannot import the LLM package — a test fails the build if it does.

LLM perimeter — choices and prose only

Local gpt-oss-20b picks among options the code allows, repairs a failed plan, writes the narrative.

Schema-validated JSON with no numeric fields; one re-ask, then the rule-based default.

Humans decide: the agent informs, proposes and waits



Cannot apply, destroy or touch AWS

no apply path in the code; plans run with dummy credentials



Cannot start itself

Gate 1: a human approves scope and budget; recorded in the ledger



Cannot merge

every change is a pull request; Gate 2 is a human review



Refuses protected resources

DR, month-end and regulatory traps are blocked before any decision

Above 100 kgCO₂e/yr the PR check fails until a human acts



Commit a suggestion

each suggestion is rendered from a template and proven with tofu plan



Or accept the increase

label eg/carbon-accepted plus a written reason; who and why go to the ledger



Below 25 kgCO₂e/yr

pass, no noise; 25–100 is a warning

Methodology: published coefficients, pinned and tested

Per resource, Cloud Carbon Footprint model

```
cpu_W = vCPU × (min + u × (max - min))  
gpu_W = GPUs × (min + u × (max - min))  
mem_W = extra GiB × memory coefficient  
kWh    = (cpu + gpu + mem) × PUE 1.135 × h  
kgCO2e = kWh × regional g/kWh ÷ 1000
```

Location-based, GHG Protocol Scope 2. Annual regional factors for totals; half-hourly UK grid data only where it changes a decision (time-shifting).

Sources

CCF coefficients vendored at commit f584c549 (Apache-2.0) · UK Carbon Intensity (CC BY 4.0) · synthetic estate, seed 42

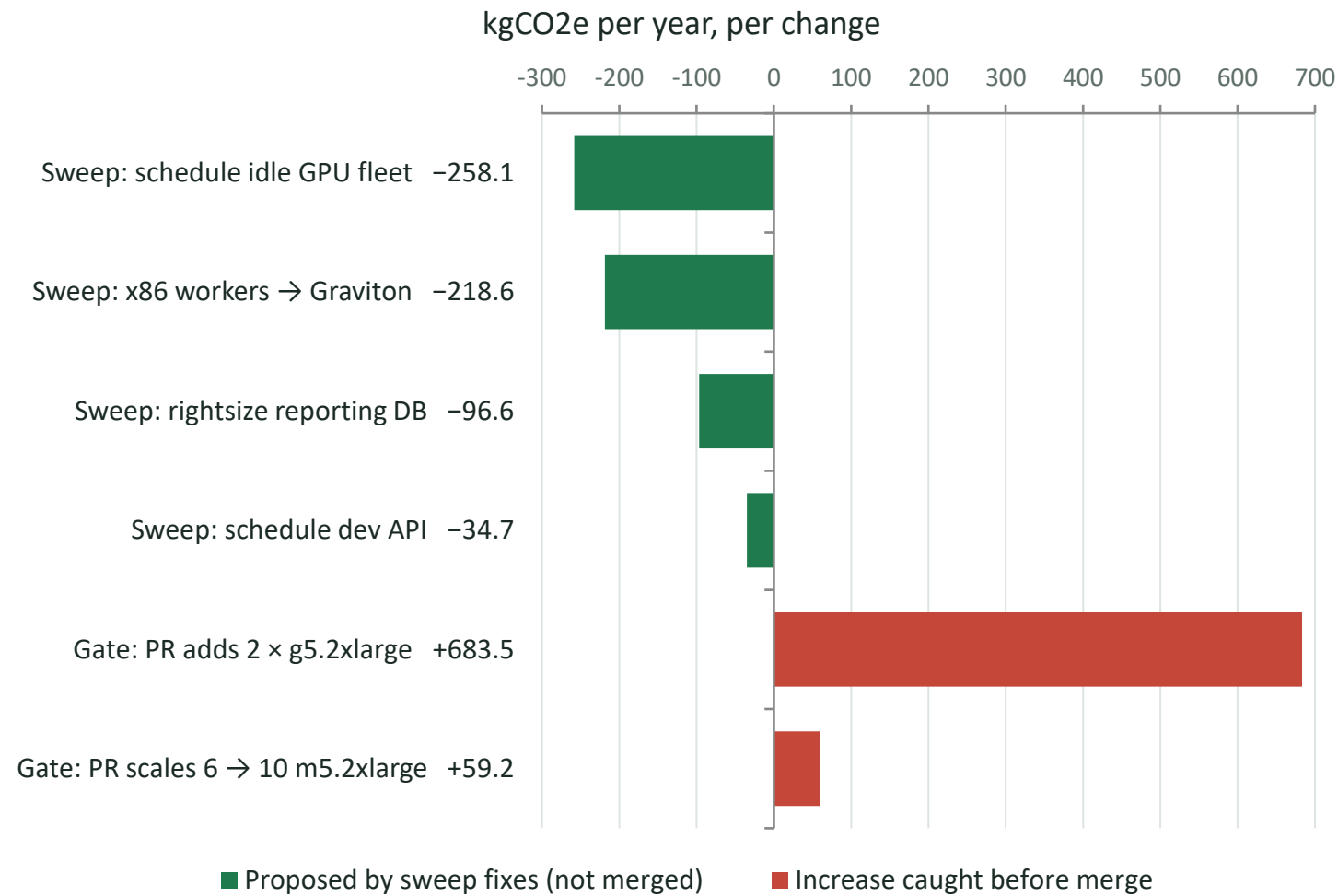
New resources are assumed

no history → CCF's 50% utilisation default, published as a 10–90% range and labelled 'assumed'

Verified, not asserted

golden values G1–G7 reproduced by tests to 3 decimals; provenance on every figure; append-only ledger

Results on the synthetic estate



608

kgCO2e/yr in four sweep fix PRs: proposed, not merged

+684

kgCO2e/yr flagged at review (range 263–1,104, assumed load)

≈ 0

kgCO2e from archiving 40 TB of logs — a cost saving, not a carbon one

Seed 42 runs: precision 1.00, recall 0.80, 0 trap violations. The agent's own footprint: 0.26 g CO2e per run offline, 3.73 g with the local LLM (estimated).

Impact: carbon becomes a review comment, not a report

	Today	With EmissionGate
When	monthly, after the fact	at merge, before the resource exists
Granularity	service and region	per resource, per change
Who sees it	a sustainability team	the engineer who can change it
What happens	a dashboard	a validated fix, or a written acknowledgement
Evidence	totals	provenance on every figure, append-only ledger



Prevention compounds

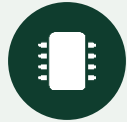
Per-change figures are tens to hundreds of kgCO2e a year — that is what the coefficients say. The gate sees every infrastructure PR, so each one caught is avoided for the resource's whole life, with no clean-up project later.

Efficiency choices: a small agent that measures itself



Deterministic maths

numbers come from code, so the model is called rarely and briefly



One local open model

gpt-oss-20b (Apache-2.0), ~3.6B active parameters, low reasoning for prose, medium for decisions



No LLM in CI

the gate is arithmetic over a plan and finishes in seconds



Templates, not generated code

fewer failed plans, fewer tokens, reviewable diffs



Hard budgets + offline mode

caps on calls, tokens, repairs, time; an LLM-free baseline to compare against



Nothing deployed

one process that runs and exits; providers from a local mirror

Own footprint, every run: CodeCarbon wraps the whole run, model included → kWh, kgCO₂e per run (SCI, ISO/IEC 21031), tokens, and payback ratio = kgCO₂e/yr proposed ÷ kgCO₂e emitted. Labelled 'estimated': CPU energy from hardware counters, RAM modelled.

Roadmap

● This submission

- Sweep, offline and with a local LLM
- PR gate in GitHub Actions with acknowledgement
- Guardrails, five fix templates, plan validation
- Own footprint in every run report

● Next

- Measured model bake-off (gpt-oss vs Qwen)
- Real fix PRs; rejection labels become rules
- Live grid API for time-shifting
- Calibrate assumed loads against what deploys
- Gate on real repos: real utilisation, credential-free plans, modules

● Later

- Azure and Google Cloud coefficients
- Same state machine on Step Functions
- Organisation-wide thresholds and reporting

Code, design document, augmentation log and demo: see the repository README.